Week 6 Cross Validation and Ridge and Lasso Regression

January 2024 Term - Quiz 2

Based on the above data, answer the given subquestions.

Consider a dataset with the following data points and the target variable:

Sample No	x	y
1	3	8
2	0	3
3	5	12
4	6	13

The linear regression model is given by $y = w_0 + w_1 x$. Assume that the Leave-One-Out Cross-Validation technique is applied.

Discussions (0)

Question 12 : 640653770615

View Parent QN | View Solutions (0) | Total Mark : 4.00 | Type : SA

Enter the value of w_1 obtained when the 3rd sample is used as the validation data point.

Answer (Numeric):

Answer

Check Answer

Solution

$$w^* = \left(XX^T\right)^{\dagger}(Xy)$$

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 0 & 6 \end{bmatrix} \; and \; y = \begin{bmatrix} 8 \\ 3 \\ 13 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 3 \\ 1 & 0 \\ 1 & 6 \end{bmatrix}$$

$$XX^T = \begin{bmatrix} 3 & 9 \\ 9 & 45 \end{bmatrix}$$

$$\left(XX^T\right)^{\dagger} = \frac{1}{54}\begin{bmatrix} 45 & -9 \\ -9 & 3 \end{bmatrix} = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{18} \end{bmatrix}$$

$$Xy = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 0 & 6 \end{bmatrix}\begin{bmatrix} 8 \\ 3 \\ 13 \end{bmatrix} = \begin{bmatrix} 24 \\ 102 \end{bmatrix}$$

$$(XX^T)^{\dagger}(Xy) = \begin{bmatrix} \frac{5}{6} & -\frac{1}{6} \\ -\frac{1}{6} & \frac{1}{18} \end{bmatrix} \begin{bmatrix} 24 \\ 102 \end{bmatrix} = \begin{bmatrix} \frac{120 - 102}{6} \\ \frac{-72 + 102}{18} \end{bmatrix} = \begin{bmatrix} 3 \\ 1.67 \end{bmatrix}$$

Question 13 : 640653770616

View Parent QN | View Solutions (1) | Total Mark : 4.00 | Type : MCQ

What will be the predicted value for the left-out data point?

OPTIONS :

- 12
- 13
- 12.3
- 11.3
- None of these

Solution

$$y = 3 + 1.67x$$
$$y = 3 + 1.67(5)$$
$$y = 11.33$$

Question 17 : 640653771205

View Solutions (0) | Total Mark : 4.00 | Type : MCQ

Given a design matrix $X \in \mathbb{R}^{d\times n}$ and a target vector $Y \in \mathbb{R}^{n\times 1}$, where d represents the number of features, n represents the number of data points, and the data is defined as:

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$Y = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

Calculate the coefficients β for Ridge regression with $\lambda = 1$.

OPTIONS :

- $\beta = [0.5, 0.5]$
- $\beta = [1, 0.5]$
- $\beta = [0.54, 0.88]$
- $\beta = [0.67, 0.33]$
- None of these

Solution

Given,

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad\qquad y = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

We Know,

$$\widehat{w}_R = \left(XX^T + \lambda I\right)^{-1} Xy$$

$$\left(\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)^{-1} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\left(\begin{bmatrix} 5 & 11 \\ 11 & 25 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right)^{-1} \begin{bmatrix} 13 \\ 29 \end{bmatrix}$$

$$\left(\begin{bmatrix} 6 & 11 \\ 11 & 26 \end{bmatrix}\right)^{-1} \begin{bmatrix} 13 \\ 29 \end{bmatrix}$$

$$\left(\frac{1}{35}\begin{bmatrix} 26 & -11 \\ -11 & 6 \end{bmatrix}\right)\begin{bmatrix} 13 \\ 29 \end{bmatrix}$$

$$\frac{1}{35}\begin{bmatrix} 19 \\ 31 \end{bmatrix}$$

$$\begin{bmatrix} 0.54 \\ 0.88 \end{bmatrix}$$

$$\therefore \beta = \begin{bmatrix} 0.54 & 0.88 \end{bmatrix}$$

May 2024 Term - Quiz 2

Question 9 : 640653853168

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

A dataset for ridge regression has 1000 data-points. k-fold cross validation is performed to estimate the value of λ in ridge regression with $k = 4$. For a given value of λ:

Discussions (0)

Question 10 : 640653853169

View Parent QN | View Solutions (0) | Total Mark : 1.50 | Type : SA

How many models have to be trained?

Answer (Numeric):

Answer

Solution

K = 4

Therefore 4 Models Will be Trained

Question 11 : 640653853170

View Parent QN | View Solutions (0) | Total Mark : 1.50 | Type : SA

While training each of these models, how many data-points does the training set have?

Answer (Numeric):

Answer

Solution

n = 1000

k = 4

Number of datapoints in each fold $= \dfrac{1000}{4} = 250$

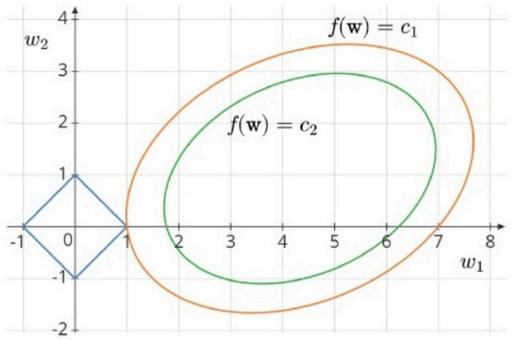
$\therefore$ While Training Models, $3 \times 250 = 750$ datapoints will be present in training set

Question 12 : 640653853171

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the following image of the parameter space in a regression problem with regularization. f is the sum of squared errors (SSE) and λ is the regularization parameter. The weight vector is $\mathbf{w} = [w_1 \quad w_2]^T$.
The contours of f and the regularization term in the loss are displayed below:



Discussions (0)

Question 13 : 640653853172

View Parent QN | View Solutions (1) | Total Mark : 1.00 | Type : MCQ

What kind of a regression problem is this?

OPTIONS :

- Ridge regression
- LASSO regression

Solution

- The blue shape in the plot represents the constraint region, which appears to be a diamond This corresponds to an L1 norm, which is used in LASSO regression.
- The ellipses represent the contours of the loss function, and the optimal solution is where the contours first touch the constraint region.
- LASSO regression applies an L1 penalty, which encourages sparsity, meaning some coefficients can become exactly zero.

Lasso Regression

Question 14 : 640653853173

View Parent QN | View Solutions (0) | Total Mark : 0.50 | Type : SA

What is the optimal value of w1?

Answer (Numeric):

Answer

Question 15 : 640653853174

View Parent QN | View Solutions (0) | Total Mark : 0.50 | Type : SA

What is the optimal value of w2?

Answer (Numeric):

Answer

Solution

The L1 constraint region (diamond shape) has its corners at points where one weight is zero (e.g)
$(t, 0)$, $(0, t)$, $(-t, 0)$, $(0, -t)$ The contour first touches the constraint at $(1, 0)$ meaning $w_1 = 1$ and $w_2 = 0$

Question 16 : 640653853175

View Parent QN | View Solutions (0) | Total Mark : 1.00 | Type : MCQ

Which of the following statements is true?

OPTIONS :

- ○ Feature-1 can be discarded.
- ○ Feature-2 can be discarded.
- ○ Any one of the two features can be discarded.
- ○ Both features are equally important and neither of them can be discarded.

Solution

Since optimal value of w_2 is 0, thus Feature 2 can be discarded

Question 17 : 640653853176

View Parent QN | View Solutions (0) | Total Mark : 2.00 | Type : MSQ

Which of the following statements are true? Exactly two options are correct.

OPTIONS :

- ☐ $c_1 > c_2$
- ☐ $c_1 < c_2$
- ☐ Total loss at optimal point is $c_1 + \lambda$
- ☐ Total loss at optimal point is $c_2 + \lambda$

Solution

The outer contour represents to $f(w) = c_1$
The inner contour represents to $f(w) = c_2$
Since contours represent levels of SSE, and the goal of regression is to minimize SSE, we know that $c_1 > c_2$

At the optimal point, The regression loss is given by the contour where the constraint is first touched. The optimal solution lies on $f(w) = c_1$ because it is the smallest attainable SSE value within the constraint.Thus, the total loss at the optimal point is : $c_1 + \lambda$

September 2024 Term - Quiz 2

Question 8 : 6406531021348

View Solutions (0) Total Mark : 4.00 | Type : SA

Consider the following dataset:

$$D = \{(1, 2), (2, 2), (0, 1)\}.$$

Assume that Leave one out cross validation is applied on this dataset and the model used is $y = w_0 + w_1 x$. Suppose $w = [w_0, w_1]^T$ be the weight obtained when $(2, 2)$ is used in the validation set. Then calculate $\|w\|_2^2$.

Answer (Numeric):

Answer

Solution

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad y = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$XX^T = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \quad \left(XX^T\right)^{\dagger} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$Xy = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$w^* = \left(XX^T\right)^{\dagger} Xy = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$||w^*||_2^2 = ||\begin{bmatrix} 1 \\ 1 \end{bmatrix}||_2^2 = \left(\sqrt{1^2 + 1^2}\right)^2 = 2$$

Question 15 : 6406531021345

Total Mark : 0.00 | Type : COMPREHENSION

Based on the above data, answer the given subquestions.

Consider the following linear regression model

$$y_i \mid x_i = w^T x_i + \epsilon_i,$$

where the noise $\epsilon \sim \text{Normal}(0, 1)$.

Discussions (0)

Question 16 : 6406531021346

View Parent QN | View Solutions (0) | Total Mark : 2.00 | Type : MCQ

For some $\lambda > 0$, where $\lambda \in \mathbb{R}$, MSE of $\hat{w}_{ML}$ is greater than MSE of $\hat{w}_{Ridge}$.

OPTIONS :

- TRUE
- FALSE

Solution

In the absence of regularization, the MLE of w is simply obtained by minimizing the sum of squared errors (SSE), which results in

$$w^* = \left(XX^T\right)^{\dagger} (Xy)$$

Ridge regression introduces an l_2 - norm penalty to the loss function

$$\widehat{w}_R = \left(XX^T + \lambda I\right)^{-1} Xy$$

The regularization parameter λ>0 reduces the variance by shrinking the weight values, potentially leading to a lower Mean Squared Error (MSE)

$$\therefore \, for\ any\ \lambda > 0\ The\ MSE\ of\ \widehat{w}_{ml} > The\ MSE\ of\ \widehat{w}_R$$

Question 17 : 6406531021347

View Parent QN | View Solutions (0) | Total Mark : 4.00 | Type : MCQ

Consider the Bayesian formulation of the linear regression problem, where the prior for w is assumed to be

$w \sim \text{Laplace}(0, 2/\lambda)$.

Then, which among the following is true?

Hint: If $X \sim \text{Laplace}(\mu, b)$, then

$f_X(x) = \frac{1}{2b} e^{-|x-\mu|}/b$.

OPTIONS :

- $\hat{w}_{MAP} = \arg\min_w \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|_2^2$, where $\|w\|_2^2 = \sum_{i=1}^{d} w_i^2$
- $\hat{w}_{MAP} = \arg\max_w \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|_2^2$, where $\|w\|_2^2 = \sum_{i=1}^{d} w_i^2$
- $\hat{w}_{MAP} = \arg\min_w \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|_1$, where $\|w\|_1 = \sum_{i=1}^{d} | w_i |$
- $\hat{w}_{MAP} = \arg\max_w \sum_{i=1}^{n} (w^T x_i - y_i)^2 + \lambda \|w\|_1$, where $\|w\|_1 = \sum_{i=1}^{d} | w_i |$

Solution

$$w \sim Laplace\left(0\,,\ \frac{2}{\lambda}\right)$$

We need to determine the correct form of the Maximum A Posteriori (MAP) Estimation for w, given this prior. The Laplace distribution has a PDF

$$f_X(x) = \frac{1}{2b} e^{\frac{-|w - \mu|}{b}}$$

Given, $w \sim Laplace\left(0\,,\ \frac{2}{\lambda}\right)$, we set $\mu = 0\ and\ b = \frac{2}{\lambda}$ so the PDF simplifies to

$$f_X(x) = \frac{1}{2 \times \frac{2}{\lambda}} e^{\frac{-|w - 0|}{\frac{2}{\lambda}}}$$

$$f_X(x) = \frac{\lambda}{4} e^{\frac{-\lambda}{2}|w|}$$

$$P(x) = \prod_{i=1}^{d} f_X(x)$$

$$P(x) = \prod_{i=1}^{d} \frac{\lambda}{4} e^{\frac{-\lambda}{2}|w|}$$

Taking Logarithm,

$$log\, P(x) \;=\; log\left(\prod_{i=1}^{d} \frac{\lambda}{4} e^{\frac{-\lambda}{2}|w|}\right) \;=\; \sum_{i=1}^{d}\left(log\, \frac{\lambda}{4} \;-\; \frac{\lambda}{2}\,|w|\right)$$

Ignoring the Constant terms,

$$log\, P(x) \;=\; \sum_{i=1}^{d} |w|$$

This is equivalent as L_1 norm

$$\|w\|_1 \;=\; \sum_{i=1}^{d} |w|$$

In linear regression, the likelihood function assumes Gaussian noise is

$$\mathfrak{L}(w\,,\, x_1\,,\, .\,.\,.\,.\,,\, x_n\,,\, y_1\,,\, .\,.\,.\,.\,,\, y_n) \;=\; \prod_{i=1}^{n} e^{\frac{-\left(w^T x_i \,-\, y_i\right)^2}{2\sigma^2}} \cdot \frac{1}{\sigma\sqrt{2\pi}}$$

$$log(\mathfrak{L}(w\,,\, x_1\,,\, .\,.\,.\,.\,,\, x_n\,,\, y_1\,,\, .\,.\,.\,.\,,\, y_n)) \;=\; \sum_{i=1}^{n} \frac{-\left(w^T x_i \,-\, y_i\right)^2}{2\sigma^2} \cdot \frac{1}{\sigma\sqrt{2\pi}}$$

$$max_w \sum_{i=1}^{n} -\left(w^T x_i \,-\, y_i\right)^2 \;\equiv\; \underset{w}{min} \sum_{i=1}^{n} \left(w^T x_i \,-\, y_i\right)^2$$

By Bayes' theorem, the posterior probability is

$$P(w \mid X, y) \;\propto\; P(y \mid X, w)\,.\; P(y)$$

Taking the log-posterior and maximizing:

$$arg\; \underset{w}{min} \sum_{i=1}^{n} \left(w^T x_i \,-\, y_i\right)^2 \;+\; \lambda\|w\|_1\,,\; where\; \|w\|_1 \;=\; \sum_{i=1}^{d} |w|$$